SAMSON TESFAMICHAEL
510.381.3854 * email : steclezion@gmail.com Oakland ,
California | Ready to Relocate

FULL STACK DEVELOPER

Motivated Full Stack PHP and PYTHON Developer with 10 years of experience in optimizing enterprise applications, backend services, and microservices architectures. Skilled in automating CI/CD pipelines, deploying with Docker and Kubernetes on AWS, and implementing microservices with Spring Boot and Kafka. Proficient in building responsive UIs with Angular, React, and modern JavaScript/TypeScript, as well as designing efficient databases with MySQL. Strong advocate of Agile practices, with a focus on streamlining development processes and enhancing performance.

TECHNICAL SKILLS

Computer Engineering | Web Programming | Artificial Intelligence | Dependency Injection
Front-End & Back-End Development | AWS | Design Patterns | SOA
Object-Oriented Programming | Database Development | Microservices | Concurrency & Multi-Threading
Networking | Data Structures & Collections | Data-Intensive Applications | JVM & Memory Management

Programming Languages: Java , JavaScript ,Python.
Web Development: Skilled in HTML5, CSS3, Jquery, Bootstrap, Ajax, JavaScript, and little Web/
App Servers: Experienced with Apache/Tomcat.
Frameworks: Strong knowledge of Spring (MVC, Boot, Security, Data, AOP, Dependency Injection, Cloud), Hibernate, Angular, Express, React, and Node.js ,Reactive native, Redux
UI-Web: SPA, JSON Restful API's, Geolocation API, HTTP Requests, Fetch API
Databases: Proficient in MySQL, MongoDB, Casandra, PostgreSQL, MS SQL, and Oracle, Laravel 12-13.x.
Backend : PHP 8.3.x , Python, Java, Go, C++, C .
Design Patterns: Singleton, Prototype, Session, Session Factory, Observer, Abstract Factory. SDLC:
Experienced in Agile/Scrum and Waterfall methodologies.
Tools: Skilled in Git, Maven, Bitbucket, Postman, Apache Kafka, ELK, Zip kin, RabbitMQ, ActiveMQ, I/CD, Docker, Kubernetes, Eclipse, IntelliJ IDEA, GitLab and GitHub.
Platforms: AWS, PCF, Azure

EDUCATION

Eritrea Institute of Technology
Bachelor of Science in Computer Engineering
(2017)

Product Architect & Lead Software Engineer **2025 – Present**
Raymoch (Hybrid)

- Conceived and designed **Raymoch**, a digital ecosystem connecting SMEs, entrepreneurs, investors, governments, and development organizations across Africa.
- Defined the end-to-end product vision, business model, subscription strategy, verification framework, and growth roadmap.
- Architected a scalable platform using Laravel, React, MySQL, REST APIs, Elasticsearch-ready search architecture, and cloud native design principles**.**
- Designed and implemented CTI (Confidence, Trust & Integrity) verification scoring, enabling business credibility assessment and risk reduction.
- Developed company intelligence modules including business profiles, financial analytics, team management, document repositories, geolocation mapping, and AI-driven insights.
- Led UX/UI design, workflow modeling, database architecture, API design, and system integration.
- Built multi-country support for regions, countries, states, cities, sectors, and industries across Africa.
- Designed investor-company matching algorithms leveraging weighted business signals, verification status, and sector intelligence.
- Created foundations for AI-powered business recommendations, market intelligence, whitespace analysis, and predictive investment opportunities.
- Defined monetization through subscription plans, premium verification services, business intelligence products, and strategic partnership programs.
- Managed product roadmap, technical leadership, stakeholder engagement, and startup execution from concept to deployment.

Startup Impact & Growth Potential

- Positioned to support millions of SMEs and entrepreneurs across African markets.
- Potential revenue streams: SaaS subscriptions, verification services, premium analytics, investor matchmaking, government partnerships, and data intelligence products.
- Designed for regional expansion into East Africa, Africa-wide markets, and global investor networks.
- Established a scalable foundation for future AI, machine learning, and predictive business intelligence capabilities.

Data Intelligence, Analytics & Visualization

- Designed enterprise-grade business intelligence architecture integrating operational, financial, verification, and market datasets into a centralized analytics ecosystem.
- Developed executive dashboards and KPI frameworks using Power BI and Tableau for real-time monitoring of company growth, investment readiness, verification status, sector performance, and geographic business distribution.
- Created interactive visualizations including heat maps, trend analysis, investment pipelines, company performance scorecards, and market opportunity dashboards using Power BI and **Tableau**.
- Designed data models and reporting frameworks supporting investor decision-making, business verification insights, and economic intelligence reporting.
- Implemented advanced analytics capabilities including predictive business scoring, sector trend analysis, startup growth forecasting, and whitespace opportunity identification.

- Developed executive-level dashboards for stakeholders, investors, government agencies, and SMEs, enabling data-driven strategic decisions.
- Built scalable reporting infrastructure capable of supporting millions of business records across multiple African countries.
- Leveraged SQL, MySQL, Laravel APIs, Power BI, and Tableau to transform complex datasets into actionable business intelligence

Technology used?

Laravel 11/12, PHP 8.x, React.js, Vite, JavaScript (ES6+), Material UI (MUI), HTML5, CSS3,RESTful APIs, JWT/Sanctum Authentication, MySQL, PostgreSQL, Elasticsearch, Meili search, SQL, Python, R Programming, ETL Pipelines, Data Warehousing, Power BI, Tableau, Business Intelligence Reporting, Interactive Dashboards, KPI Scorecards, Data Visualization, Predictive Analytics, Machine Learning, TensorFlow, PyTorch, Google Maps API, Geospatial Analytics, CTI (Confidence, Trust & Integrity) Verification Framework, Investor Matching Algorithms, Business Intelligence Platforms, Cloud Computing (AWS/Azure), Docker, Git, GitHub, Agile Scrum, System Architecture Design, Database Modeling, API Development, UI/UX Design, Market Intelligence, Business Verification Systems, Financial Analytics, Document Management Systems, AI-Driven Insights, Business Opportunity Mapping, Risk Scoring Models, Startup Ecosystem Development, and Data-Driven Decision Support Systems. Interactive dashboards, Power BI, Tableau, and data visualization capabilities are recognized core skills for modern analytics and business intelligence platforms.

Senior Software Engineer
Nile IT Solutions
(Remote) ***Oct 2024 – present***

- Implemented Agile/Scrum practices to manage the full Software Development Life Cycle(SDLC) for enterprise Laravel-based and Python-based backend applications.
- Designed and developed RESTful APIs using Laravel 11.x and Python frameworks (Django, leveraging Eloquent ORM, Service Providers, Repository patterns, and modular architecture principles to build scalable, maintainable systems.
- Built and maintained secure API authentication and authorization layers using Laravel Sanctum, JWT, middleware guards, and Python-based security modules.
- Demonstrated strong proficiency as a Backend / Python Software Engineer, developing microservices, automation workflows, asynchronous task queues, and intelligent Python Agent services for data processing, orchestration, and system optimization.
- Developed automated test suites using PHP Unit (Laravel Feature & Unit Tests) and integrated Jest **and** Python testing frameworks (PyTest / unittest) for front-end, API contract, and backend validation to ensure code reliability and coverage.

- Migrated legacy Python mock testing servers into structured Laravel and Python testing environments using built-in HTTP testing utilities, service mocking, and dependency injection strategies.
- Implemented centralized logging and monitoring through Laravel Monolog, Python logging modules, debugger dashboards, and Splunk indexing/queries to track real-time application health and performance.
- Enforced strict code quality standards through Laravel and Python best practices, reusable service classes, traits, modular packages, and consistent architectural patterns.
- Managed CI/CD pipelines using GitHub Actions/Pipelines, and automated build-test-deploy workflows for Laravel and Python backend services.
- Produced technical system architecture documentation, API specifications (OpenAPI/Swagger),and lifecycle reports to support stakeholder communication and sprint reviews.
- Containerized Laravel and Python applications using Docker**,** created optimized Dockerfiles and docker-compose workflows, and orchestrated deployments using Kubernetes for scalable, cloudready infrastructure.

Technologies Used

Laravel 11.3, PHP 8.4, Eloquent ORM, RESTful API Architecture, JavaScript , Jest (for front-end/unit testing), PHPUnit (Laravel Unit & Feature Testing, Jest (JavaScript testing framework), Mocking and API test environments, MySQL / MariaDB, Database Migrations & Seeders, Git, GitHub, Bitbucket, GitHub Pipelines / Bitbucket Pipelines, CI/CD Automation, Laravel Monolog, Splunk (log indexing and queries), Debugging Dashboards, Docker, Docker Compose, Kubernetes, Agile / Scrum, API Documentation (Swagger / OpenAPI, Microservices Architecture.

Software Engineer / Team Leader
Human Resource & Fraud Detection Systems , Office of the Governor (OOG)(Ministry of Local Government ,
Eritrea, Asmara **April 2020 – Oct 2022**
(In-person)

- Led and collaborated with cross-functional development teams to design and integrate enterprise scale fraud detection components into core HR and financial systems, ensuring secure data flow, real-time transaction monitoring, and audit compliance.
- Architected and maintained RESTful APIs and web services using Spring Boot (Java) and Laravel 8.x (PHP), delivering scalable, high-performance service-oriented architectures deployed across cloud and on-prem environments.|

- Designed and modernized legacy enterprise systems, replacing monolithic and outdated components with microservices-based architectures, modular service layers, and cloud-native patterns to improve scalability and maintainability.
- Deployed and operated serverless Laravel applications using Laravel Vapor, leveraging AWS Lambda, SQS, CloudWatch, and RDS to achieve high availability, automated scaling, and reduced operational overhead.
- Integrated Elasticsearch to power advanced search, analytics, and fraud-detection indexing, enabling near real-time querying, anomaly detection, and fast retrieval across large HR and financial datasets.
- Developed and managed SOAP Request/Response services, performed API testing using Postman, and implemented robust validation, authentication, and authorization frameworks.
- Built enterprise-level Human Resource Management Systems (HRMS) covering recruitment, performance management, workforce planning, payroll, compensation, succession planning, and employee training modules.
- Implemented real-time system features including WebSockets, push notifications, alerts, and reminders using Laravel broadcasting, event-driven architecture, and queue-based background processing.
- Designed, optimized, and continuously tuned fraud detection algorithms in Java, improving detection accuracy, throughput, and system reliability through profiling, benchmarking, and iterative refinement.
- Designed and maintained Splunk dashboards for centralized logging, real-time monitoring, query indexing, security auditing, and application performance analysis across distributed systems.
- Performed advanced database design and optimization using Oracle, MySQL, PostgreSQL, and SQLite, including schema design, triggers, views, stored procedures, and performance-tuned SQL.
- Developed complex PL/SQL packages, procedures, and functions, achieving measurable improvements in data processing efficiency, transaction consistency, and overall system stability. • Implemented and executed automated testing pipelines using Cucumber (BDD), Selenium, TestNG, and PHPUnit, ensuring high-quality code delivery, regression prevention, and continuous integration reliability.
- Diagnosed and resolved production-critical issues through proactive troubleshooting, root-cause analysis, and live system debugging, maintaining high availability and minimizing downtime.
- Practiced Agile/Scrum methodologies, contributing to sprint planning, backlog grooming, code reviews, release management, and continuous improvement initiatives.
- Built and maintained CI/CD pipelines using GitHub, Jenkins, and Pivotal Cloud Foundry (PCF), automating build, test, and deployment workflows across development, staging, and production environments.

Technologies Used

Java, Spring Boot, Spring MVC, Laravel 8.x
RESTful APIs, SOAP Web Services, Microservices Architecture, JavaScript, React, Vue, HTML5, CSS3, Oracle Database, MySQL, PostgreSQL, SQLite, PL/SQL, Triggers, Views, Stored Procedures, Cucumber (BDD), Selenium, TestNG, PHP-Unit, Automation Testing, GitHub, Jenkins, CI/CD Pipelines, PCF (Pivotal Cloud Foundry), Splunk, Kafka, Agile, Scrum

Software Engineer / Team Leader – Enterprise Archive & Records Management Systems RCC & Office of the Auditor General (OAG)
Asmara, Eritrea **Jun 2019 – Mar 2021**
(In-person)

- Collaborated as a core member of agile software development teams, contributing across the full Software Development Life Cycle (SDLC) including system design, development, testing, deployment, and long-term maintenance of mission-critical enterprise applications.
- Performed detailed business and technical requirement analysis, translating functional requirements into scalable system architectures and technical specifications to align development with operational goals.
- Led the design and implementation of enterprise archive and records management systems**,** supporting secure capturing of documents, emails, and digital files from multiple predefined data sources.
- Architected and developed RESTful web services following industry best practices, using layered and microservices-based architectures to ensure system scalability, reliability, and maintainability.
- Designed and implemented real-time record circulation workflows, automating document tracking, approvals, transfers, and lifecycle management based on defined operational flows.
- Built advanced multi-criteria search engines enabling rapid retrieval of records based on document metadata, archive boxes, classification tags, and physical location references.
- Led the development team at the Office of the Auditor General, providing technical leadership, code reviews, architecture guidance, and delivery oversight to ensure high-quality output.
- Practiced Test-Driven Development (TDD) principles to improve software reliability, maintainability, and early defect detection during the development cycle.
- Optimized database architectures using MySQL and Oracle, implementing triggers, views, indexes, and stored procedures to enhance system performance and data integrity.
- Ensured strict adherence to coding standards, documentation practices, and repeatable implementation patterns across projects.

Technologies Used
Laravel 8.x, PHP, RESTful Web Services, Microservices Architecture, JavaScript, React, Vue.js, HTML5, CSS3, MySQL, Oracle Database, Stored Procedures, Triggers, Views ,TestNG, Automated Testing, TDD Practices, GitHub, PCF (Pivotal Cloud Foundry).

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HR Analyst / Data Engineer / Team Leader
Tourism Management Systems
Resource Coordinating Center (RCC) & Ministry of Tourism,
Asmara, Eritrea **Aug 2018 – Feb 2020**
(In-person)

- Applied statistical modeling and workforce analytics to evaluate employee-related factors, enabling data-driven decision-making to improve talent acquisition, performance management, employee retention, and organizational development.
- Designed and delivered HR analytics dashboards and reporting frameworks**,** supporting strategic workforce planning and operational improvements for leadership stakeholders.

- Led and participated in data engineering and data collection initiatives**,** developing scalable data pipelines using Python, R, SQL, MySQL, and PHP following industry best practices.
- Conducted solution architecture reviews and technical assessments in collaboration with multiple cross-functional business and technology teams, contributing to the selection and evaluation of new development tools and platforms.
- Worked closely with information security teams to evaluate and approve over 1,000+ Python and R libraries and open-source tools, ensuring security compliance and operational integrity.
- Received structured mentorship from senior developers, accelerating technical proficiency and leadership capabilities through hands-on enterprise development exposure.
- Led the design and development of an Enterprise Tourism Management System**,** providing end-toend digital solutions for reservation and booking, CRM, inventory management, accounting, billing, reporting, and business analytics.
- Architected and implemented web-based systems using modern front-end technologies including HTML5, CSS3, JavaScript, Dojo, XML, and XSLT, delivering interactive and responsive user experiences.
- Built secure, scalable backend services using PHP, MySQL, and SQL Server**,** ensuring high system performance, data consistency, and audit-ready reporting.
- Managed system workflows, user roles, and access control to support operational efficiency and regulatory compliance.

Technologies Used

Python, R, PHP, SQL, JavaScript, RESTful Web Services, HTML5, CSS3, JavaScript, Dojo Toolkit, XML, XSLT, MySQL, SQL Server, Open-source library security evaluation, Information security best practices, Data-driven HR analytics, Agile collaboration and SDLC practices.

HR Analyst / Data Engineer / Team Leader – Censuses & Demographic Systems

Resource Coordinating Center (RCC) & Ministry of Local Government
Asmara, Eritrea **Dec 2017 – Sep 2019**
(In-person)

- Applied statistical modeling and workforce analytics to evaluate employee-related factors, enabling data-driven decision-making in talent acquisition, retention, performance evaluation, and organizational development.
- Designed and delivered HR analytics dashboards and automated reporting frameworks to support strategic workforce planning and improve operational transparency for senior leadership.

- Led and participated in large-scale data engineering initiatives, developing scalable ETL pipelines using Python, R, SQL, MySQL, and PHP, ensuring accuracy, consistency, and compliance across datasets.
- Conducted solution architecture reviews and technical assessments with cross-functional business and technology teams, contributing to the evaluation and adoption of new development tools and platforms.
- Collaborated with information security teams to evaluate and approve 1,000+ open-source Python and R libraries, ensuring regulatory compliance and secure ecosystem integration.
- Received structured mentorship from senior developers, enhancing technical depth, architectural thinking, and leadership capabilities through hands-on enterprise development exposure.
- Led the design and development of an Enterprise Tourism Management System, enabling digital transformation across reservation & booking, CRM, inventory, accounting, billing, reporting, and analytics modules.
- Architected and implemented interactive web-based systems using HTML5, CSS3, JavaScript, Dojo, XML, XSLT**,** delivering responsive, user-friendly interfaces for operational teams.
- Built secure, scalable backend services in PHP, MySQL, and SQL Server, ensuring high system performance, data consistency, and audit-ready reporting capabilities.
- Managed workflows, user roles, and access control models to support operational efficiency and regulatory compliance.

Major In-Person Data Engineering Project (Ministry of Local Government)

- Worked as a Software Engineer on a joint RCC–Ministry of Local Government initiative focused on collecting, validating, and refining demographic data from multiple regions.
- Processed highly unrefined and inconsistent demographic datasets collected without proper validation rules by designing a robust, multi-layered data cleansing and transformation pipeline**.**
- Converted legacy MS-SQL datasets into SQLite using Python, Flask, SQLAlchemy, and a suite of 14+ supporting Python libraries, ensuring structural integrity and normalized schema design.
- Designed and implemented a sophisticated validation engine capable of validating over 20 critical demographic data columns**,** including identity fields, geographic markers, age classification, and household metadata.
- Successfully processed and transformed over 150 million records within four days using a Core i9 workstation, achieving near-zero data loss and producing fully refined, analysis-ready datasets.
- Utilized key dependencies including PyQt-Python, SQLAlchemy, Flask**,** and additional ETLfocused libraries to automate conversion, cleaning, error detection, and logging.
- Followed Agile–Scrum methodology**,** conducting daily standups, sprint planning, and iterative validation review cycles with data officers, statisticians, and technical leads.

Technologies Used

Python, Flask, SQLAlchemy, PyQt-Python, R, PHP, SQL, JavaScript, RESTful Web Services, HTML5, CSS3, JavaScript (Dojo Toolkit), XML, XSLT, MySQL, SQL Server, Data validation frameworks, ETL/ELT pipelines, Open-source library security evaluation, Agile/Scrum, HR analytics, Information security best practices.

Software Developer / Data Engineer /

Queue Management System – Payroll & HR Automation (Ministry of Finance)

Eritrea, Asmara **Aug 2017 – Oct 2022**

(In-person)

Developed an enterprise-level Queue Management System used by the Payroll Department, Human Resource Department, and the Resource Coordinating Center of the Ministry of Finance, providing a secure digital workflow for government organizations to submit payroll requests, process employee data, and ensure real-time financial approvals.

Key Contributions & Responsibilities

- Engineered a secure, multi-tenant authentication system where every organization signs in using a unique Org ID and Class ID, with session management, cookie validation, and login approval protocols tailored for government workflows.
- Built real-time communication and approval channels connecting organizational managers**,** HR staff**,** and financial officers, including OTP verification and dual-department approval mechanisms**.**
- Integrated a dedicated Reconciliation System to cross-check newly uploaded payroll documents against historical records, ensuring accuracy, fraud prevention, and audit compliance.
- Leveraged Python and R libraries for complex mathematical and statistical analysis, supporting payroll calculations, anomaly detection, timesheet validation, overtime and deductions, and historical pattern analysis.
- Designed and developed a modern, responsive frontend using React Native and Angular 2, with advanced UI behaviors such as spinners, progress indicators, multi-stage loaders, and data-driven transitions to handle heavy background processing.
- Built backend services using PHP 7.4, integrating with Python/R analytics modules for the computation phase, enabling a hybrid architecture optimized for performance and accuracy.
- Implemented an automated payroll comparison engine where requested amounts are validated against previous payroll entries, employee status, timesheets, overtime records, leave types (annual/monthly), and deduction rules before generating the final payable amount.
- Designed system components that automatically calculate payroll totals and route approved payments to individual employee bank accounts, ensuring end-to-end automation.
- Delivered advanced progress bars, analysis dashboards, and real-time charts to visualize processing stages, organizational performance, and financial analytics metrics.
- Produced advanced reporting dashboards for executive leadership using Power BI and Tableau, enabling pivot analysis, multi-variable comparisons, and OOP-structured report generation for higher-level decision makers.

Technologies Used

PHP 7.4**,** React Native**,** Angular 2**,** Python**,** R**,** SQL Server**,** MySQL**,** REST APIs**,** XML/JSON**,** Power BI**,** Tableau**,** Session & Cookie Authentication**,** Queue Processing**,** Real-time Workflows**,** Data Reconciliation**,** ETL/Validation Pipelines**,** Statistical Modeling**,** Government Payroll Automation